Experiment-19

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Course Code: -MLA0305 **Slot-**B

Title: Reinforcement Learning in Partially Observable Markov Decision Processes

Aim:
To implement a Reinforcement Learning agent for solving POMDP environments and evaluate decision-making under uncertainty.

Procedure:

1. Define a partially observable environment.
2. Define the hidden states and available observations.
3. Allow the agent to receive incomplete observations.
4. Maintain a history or belief representation of previous observations.
5. Select actions based on the available information.
6. Execute actions and receive rewards.
7. Update the agent's policy using the observed experience.
8. Repeat the process for multiple episodes.
9. Measure cumulative reward and decision accuracy.
10. Analyze the effect of partial observability.

Result:

The RL agent successfully learned to make decisions using incomplete observations. The experiment demonstrated how Reinforcement Learning can handle uncertainty in POMDP environments.